

VIBEGUARD - INTERIM FIRMWARE SPECIFICATION

Hardware Bring-Up, Real-Time DSP Telemetry, Demonstration Scope & Anti-Scope Specification

DOCUMENT ID	FIRMWARE REVISION	TARGET MCU	SENSOR PROTOCOL	AUTHOR / TEAM
VG-FW-INT-01	v1.0-INTERIM-DEMO	7Semi ESP32-DEVKIT-E	ADXL345 SPI Mode 2	VibeGuard Project Team

1. Executive Summary & Context of Existence

Purpose: This specification establishes the technical baseline, operating principles, and rigorous operational boundaries for the interim demonstration firmware (VibeGuard_Presentation_Demo.ino). The software has been compiled and burned directly into the onboard non-volatile SPI flash memory (address 0x00010000) of the 7Semi ESP32-DEVKIT-E microcontroller to support the Semester First Review presentation.

Because custom 3D-printed mechanical components (the FAB-01 eccentric rotor arm and FAB-02 rigid sensor bracket) are presently undergoing fabrication in the college FabLab, this firmware serves as a physical proof-of-concept. It bridges bare-metal hardware assembly with mathematical signal processing, proving genuine silicon sensor interfacing, rigorous electrical galvanic isolation, dynamic gravity DC bias removal, and ISO 10816-3 severity classification on real hardware without waiting for finished mechanical tooling.

2. What the Program IS Doing (Active Implementation Details)

Active Functions: The flashed firmware is actively executing the following verifiable physical and digital operations in real time:

- 1. Certified SPI Bus Interfacing (Mode 2):** Communicates with the ADXL345 3-axis accelerometer over a 1 MHz 4-wire hardware SPI bus (GPIO 18 SCK, GPIO 19 MISO, GPIO 23 MOSI, GPIO 21 CS). Rigorous hardware audits verified that SPI Mode 2 (CPOL=1, CPHA=0) achieves 100% bit-perfect register write and read-back latching on this breakout module.
- 2. Genuine Silicon Verification:** Queries register 0x00 at startup and halts until the authentic Analog Devices silicon identifier DEVID 0xE5 is received, preventing execution against disconnected or floating buses.
- 3. High-Rate Digital Acquisition:** Configures the ADXL345 for 800 Hz Output Data Rate (BW_RATE = 0x0D, Nyquist frequency = 400 Hz) and Full-Resolution +/-16g measurement range (DATA_FORMAT = 0x0B, 3.9 mg/LSB scale factor).
- 4. Synchronous Burst Transfers:** Gathers real acceleration data in 128-sample windows (160 ms per analysis window) using multibyte burst reads (register 0x32 with Read [0x80] and Multibyte [0x40] bits asserted), completely eliminating register skew.
- 5. Dynamic DC Gravity Elimination:** Dynamically calculates window mean acceleration across all 3 axes (X, Y, Z) and subtracts it from every instantaneous sample ($dx = x - \text{mean}_x$). This cleanly filters out the static 1.0g Earth gravity vector and physical bench tilt, isolating pure dynamic AC mechanical vibration.
- 6. Dynamic Vector RMS Calculation:** Computes the true 3-axis geometric magnitude Root Mean Square (V_{RMS}) across the 128-point sample window using double-precision float Euclidean summation.
- 7. ISO 10816-3 Decision Engine:** Evaluates V_{RMS} against an alarm threshold of 0.350g (boundary between ISO Class I/II acceptable vibration and unacceptable severity). Incorporates a 2-stage persistence debounce filter (K-of-M counter) to prevent transient mechanical bench noise from causing nuisance trips.
- 8. Autonomous Standalone Operation:** Executes immediately upon receiving 5V USB power (from laptop, power bank, or phone charger). Powers the onboard tri-color LED solid GREEN for Good Health / Normal baseline, and switches to solid RED under anomaly conditions.
- 9. Dual Telemetry Streaming:** Outputs both high-readability formatted diagnostic logs and high-speed Arduino Serial Plotter datagrams (>VRMS:...,AlarmThresh:...,State:...) over /dev/ttyUSB0 at 115200 baud.

3. What the Program IS SUPPOSED To Do (Review Objectives)

Evaluation Criteria: For tomorrow's academic and faculty evaluation, the firmware is specifically tasked with meeting four core demonstration objectives:

- **Objective A - Live Sensor & DSP Proof:** Demonstrate to reviewers that the team has successfully brought up real physical sensors, established clean high-speed SPI communication, and validated real-time mathematical vibration DSP on an embedded 32-bit RTOS architecture.
- **Objective B - Healthy Baseline Verification:** Show that under normal resting conditions or decoupled motor spinning, the baseline vibration rests at $V_{RMS} \sim 0.017g$, well within the ISO 10816-3 permissible envelope with a solid GREEN indicator.
- **Objective C - Multi-Channel Fault Injection:** Provide interactive, verifiable means to simulate mechanical unbalance and trigger the ALARM condition (Solid RED LED + telemetry warning) without requiring physical unbalance weights (see Section 6).
- **Objective D - Electrical Domain Isolation:** Demonstrate that the 12V 2A motor drive circuit is 100% galvanically isolated from the sensitive 3.3V ESP32 logic, preventing inductive flyback spikes or brownout resets.

4. What the Program IS NOT (Explicit Anti-Scope)

Scope Boundaries: To ensure complete academic transparency and prevent misconceptions, the following boundaries define what this interim program is explicitly NOT doing:

CRITICAL BOUNDARIES & EXPLICIT ANTI-SCOPE (WHAT THIS IS NOT)

- **NOT the Frozen Production Firmware:** This code is strictly an isolated scratch demonstrator (VibeGuard_Presentation_Demo.ino). It does NOT alter, overwrite, or merge into the frozen master firmware directory (07_SEMESTER_EXECUTION/03_Firmware_and_Simulation/VibeGuard_ESP32_Firmware/).
- **NOT Physical Mass Unbalance:** Because the FAB-01 eccentric rotor arm is not yet installed on the N20 motor shaft, the current bad health state is demonstrated via interactive mechanical tap or hardware button injection, NOT centrifugal mass displacement.
- **NOT Full FFT Spectral Decomposition:** This demonstrator evaluates statistical time-domain Vector RMS. Full 512-point Fast Fourier Transform (FFT) for isolating 1X rotational frequency and harmonic bearing peaks is scheduled for Phase 4 after rigid mounting.
- **NOT Cloud / MQTT / IoT Connected:** Wi-Fi transmission and MQTT cloud publishing are deliberately held dormant to prevent RF power bursts (500 mA) and ensure 100% stable, zero-latency execution on the podium without college Wi-Fi login issues.
- **NOT an Electrical Diagnostic Tool:** It measures mechanical acceleration only; it does not sense motor armature current, winding thermal dissipation, or supply voltage ripple.
- **NOT a Synthetic Mock / PC Simulation:** Unlike earlier Python test scripts, every single acceleration bit is physically acquired in real time from genuine ADXL345 MEMS silicon over the physical SPI bus.

Engineering Rigor Note: These boundaries guarantee that tomorrow's presentation accurately portrays current semester progress: 100% physical bring-up completed, DSP verified, and software architecture validated, awaiting final 3D mechanical press-fit parts for full unbalance characterization.

5. Hardware Pinout & Wiring Interface Reference

Interface Mapping: The following pinout reflects the physical bench assembly verified and active on the hardware carrier:

SIGNAL	ESP32 PIN	BREADBOARD	ADXL345 PIN	ELECTRICAL DOMAIN	FUNCTION
GND	GND	Blue Rail (-)	Pin 1 (GND)	3.3V Logic Ground	Common reference
VCC	3.3V Out	Red Rail (+)	Pin 2 (VCC)	3.3V Regulated (AMS1117)	Sensor supply (140 uA)
CS	GPIO 21	Col 22, Row F	Pin 3 (CS)	3.3V CMOS Logic	SPI Chip Select (Active Low)
SDO	GPIO 19	Col 25, Row F	Pin 6 (SDO)	3.3V CMOS Logic	SPI MISO (Master In Slave Out)
SDA	GPIO 23	Col 26, Row F	Pin 7 (SDA)	3.3V CMOS Logic	SPI MOSI (Master Out Slave In)
SCL	GPIO 18	Col 27, Row F	Pin 8 (SCL)	3.3V CMOS Logic	SPI SCLK (1 MHz Mode 2)
LED GREEN	GPIO 25	Col 12, Row F	Cathode (330 Ohm)	3.3V PWM / Digital	Normal State Indicator
LED BLUE	GPIO 26	Col 14, Row F	Cathode (330 Ohm)	3.3V PWM / Digital	Boot Self-Check Indicator
LED RED	GPIO 27	Col 15, Row F	Cathode (330 Ohm)	3.3V PWM / Digital	Alarm State Indicator
BOOT BTN	GPIO 0	Onboard Button	N/A	Internal Pull-Up	Interactive Fault Injection
MOTOR +	External 12V	Screw Terminal	N20 (+) Wire	12V 2A Isolated Domain	Isolated Drive Rail
MOTOR -	External GND	Rocker / Fuse	N20 (-) Wire	12V 2A Isolated Domain	1N4007 Clamped Return

6. Presentation Demonstration Script & Operator Guide

Demonstration Flow: Follow this exact step-by-step procedure during tomorrow's presentation to demonstrate complete system capability:

PHASE	OPERATOR ACTION	TELEMETRY OUTPUT	VISUAL LED	REVIEWER TAKEAWAY
1. Cold Boot	Plug 5V USB cable into ESP32	[SENSOR] ADXL345 DEVID: 0xE5 ... VERIFIED POWER_CTL=0x08, DATA_FORMAT=0x0B	Solid BLUE (0.5s) then Solid GREEN	MCU boots from internal flash; verifies silicon over Mode 2 SPI.
2. Baseline Good Health	Motor running or bench resting	>VRMS:0.017,AlarmThresh:0.35,State:0 Status: [NORMAL / GOOD HEALTH]	Solid GREEN	Proves dynamic DC gravity removal and smooth Class I/II baseline.
3. Physical Vibration Tap	Gently tap bench near sensor	>VRMS:0.420,AlarmThresh:0.35,State:1 Status: [ALARM / UNHEALTHY]	Solid RED	Proves physical dynamic sensing and ISO 10816-3 thresholding.
4. Electronic Fault Injection	Press & hold ESP32 BOOT button (GPIO 0)	>VRMS:0.719,AlarmThresh:0.35,State:1 Status: [ALARM / UNHEALTHY] <-- [BOOT]	Solid RED	Demonstrates simulated rotor unbalance (+0.70g) without 3D parts.
5. Recovery	Release BOOT button	>VRMS:0.018,AlarmThresh:0.35,State:0 Status: [NORMAL / GOOD HEALTH]	Solid GREEN	Validates 2-stage persistence debounce preventing nuisance trips.
6. Serial Control (Optional)	Type 'a' or 'n' in Serial Monitor	>>> [SERIAL COMMAND] FORCED FAULT INJECTED / CLEARED <<<	RED on 'a' GREEN on 'n'	Demonstrates interactive supervisory command interface.